

Institute of Actuaries of India

Subject SA7-Investment and Finance

March 2022 Examination

INDICATIVE SOLUTION

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Listed debt**

- Listed debt is sold on an exchange
- Both government bonds as well as corporate bonds and asset backed securities can be listed on exchange

Advantages

- The price is more readily available
- Theoretically, liquidity is expected to be higher than for private debt – but listed debt is not always liquid and often seldom trades;
- Mainly because most of the issues are held by institutional investors to match their longer term liabilities and are held till maturity; Debt securities may be a match or reference for institutional liabilities, accounting or capital assessments
- Buying/selling decisions can be quickly made and potentially much cheaper than private debt (compared to the potential cost of forced selling of private debt)
- Listed debt has stricter regulatory / compliance requirements
- Ratings of debt help establish individual bond issuer creditworthiness
- Investors may be limited to investment grade credit or other rating based restrictions
- Future funding requirements for borrower can be met from the broader marketplace, not just the existing investors
- Within listed debt diversification can be more easily achieved as there are comparatively more investment opportunities (although have reduced in recent years and issuers may concentrated in certain sectors)
- In some advanced markets, derivatives based on listed debt can be purchased. This can potentially help the manager with asset allocation as well as limiting negative returns (at a cost)
- Higher allocation can be made according to regulation
- Typically number of issues and their depth is larger for listed debt

Private debt

- Private debt is owned by a private group of investors, usually including the owners, management of the company issuing the debt or a syndicate of investors providing debt based funding.
- Mostly corporate bonds that can be held privately; in some extreme cases Govt securities can be issued only to specific investors with caveats around offloading/ selling to other parties.

Advantages

- Private debt provides diversification benefits to other asset classes
- These bonds are have less market pricing efficiency; thus offering the potential of higher returns to compensate for lower liquidity
- Liquidity premium will also increase returns compared to listed
- Higher degree of manager influence and governance – can force borrower to agree on stricter covenants
- Credit worthiness of the issuer may not available and even if its available, the nature of covenants may not be available to determine the credit worthiness of the issue, investors have to perform their own due diligence in the absence of credit rating of the issue.
- Will be difficult to hedge against market risk

[7]

ii)

a)

liquidity risk – an investment into private debt cannot easily be liquidated and term of lending is subject to prepayment risk ○ if an opportunity is unwound, monies will be distributed back to investors ○ Similarly, recurring debt repayments may be reinvested or paid to investors	○ Distribute proceeds of debt repayment directly to investors. ○ Structure the fund in a way that loans can be given against the investment if liquidity by investors is required; albeit at some margin ○ Set minimum lockin period and associated penalties upfront for investors. ○ Set rules regarding foreclosure/ recurring repayment ○ Set framework to recover as much amount as possible in case of default
Currency risk – consideration needs to be given whether or not currency risk should be included. ○ If included it is expected to increase the returns (due to inflation differential), however it does increase the risk. ○ If target market's liabilities for investor are USD denominated, then currency risk should probably be hedged out. Though, this will increase the cost of the fund.	○ The overall limit for investors to invest overseas needs to be considered ○ Set % in unlisted debt instruments for retirement funds. This limit may reduce the potential to raise the capital for the fund.
Default risk – high risk of default beyond the risk appetite of the investor ○ Smaller position as part of the investor's overall fund can be made ○ Could potentially target lower risk of default companies	○ Is the return target sufficient compensation for the risk taken? ○ Obtain credit ratings for further comfort)), or specify limits on higher risk debt; however, this might be to the detriment of returns.
Risk return profile: Is the return target sufficient compensation for the risk taken?	○ What returns have other funds similar to this one produced over the past ○ How will the market differ in the foreseeable future (for example when QE ends), and what influence will that have on the market the portfolio manager is operating in?
Fees: Are the fees too high – there has been focus on fees for institutional investors	○ For the target market a detailed analysis of the fees will need to be provided – show total fees payable under different performance outcomes ○ How does the 1.5% compare to similar products and how was this determined? The 8% hurdle needs to be assessed against other funds
Operational and regulatory risk: The manager will have to prove it has the necessary processes in place to ensure that the fund is audited and complies with all regulation	○ Audit and compliance functions need to be in place and evidence of robust management of operational risks including disaster recovery ○ Ideally this may be outsourced to specialist agency for cost efficiency and in order to give investors peace of mind
Tax implications: What will the tax implication be for the target clients?	○ Design the fund be structured in a more tax effective manner considering minimum lock in, sectoral exposure etc that make fund eligible for favourable taxation ○ The tax implication of an international fund need to

	be clearly understood and communicated – i.e. double tax relief, who reclaims withholding tax, how will Indian/ US tax exempt investors be handled. ○ The tax implications depend on the type of institutional investor.
Legal Structure of the fund	○ Set the legal structure of the fund that is tax efficient and does not dissuade potential investor to invest in the fund. ○ e.g. can buy in as a limited partner which means that they can only lose as much as they've put in. ○ If structured as a company, then there will be company tax payable before the distribution of profits/income to investors.

[7]

b)

The capital commitment phase is typical and should not be perceived abnormally low by the target market	○ A discount on fees to investors who commit capital first. This will also help with the marketing angle of 'buy now before it is too late'
The \$10 m might be considered too little compared to the intended size of the fund	○ Either consider reducing the size of the overall fund, or increase the capital commitment. This will show the potential investors that the PM's interest is aligned to that of the investors. ○ Commit higher initial capital with drawdown as the fund reaches threshold limit of say \$100m
The reputational risk that the investors don't understand the product which later causes reputational damage	○ This fund should only be marketed to sophisticated institutional investors due to the financial knowledge required as well as risk inherent in this fund. ○ High minimum investment needs to be introduced to deter smaller investors from accessing this fund ○ Marketing material must explicitly explain all investment risks
Will the fund achieve the critical minimum inflow that has to be achieved in order for the fund to be optimally functioning?	○ It would be ideal if the capital committed by the fund managers covers the critical minimum. ○ Commitments could be obtained from a few key investors before going to market. ○ Widen the net of potential investors e.g. beyond institution or outside of India
Will the manager be able to sell his product if he is unknown in the Indian market?	○ A list of the fund manager's past experience and track record needs to be included in the marketing material as well as communicated to the potential investments (pipeline). ○ The marketing material should include why is this asset class may be potentially more attractive than other investments ○ Marketing material should also include the maximum investment in an underlying opportunity in order to achieve diversification and reduce concentration risk. This will be a key risk in the beginning of the fund when only a few investments have been made. ○ The fund manager's past performance could be used to show

	diversification benefit if combined with typical traditional strategies ○ The manager needs to offer something special (unique selling point) for investors to invest in an unknown manager.
Audit and compliance functions need to be in place with robust management of operational risks including disaster recovery.	○ Audit and compliance functions need to be in place and evidence of robust management of operational risks including disaster recovery ○ Ideally this may be outsourced to specialist agency for cost efficiency and in order to give investors peace of mind
Timing of launch will be important	○ December is generally bad month to launch as most investors are on holiday, March and June is most people's year end thus too busy to look at exotic investments
Lower return than expected	○ May not get the anticipated level of returns (i.e. above 8%) which could result in low fees collected and costs not being covered

[8]

- iii) The Effective Annual Cost measure (EAC) is a standardised disclosure methodology that can be used by consumers and advisors to compare charges on most retail investment products, and their impact on investment returns, so that consumers are placed in a position to make better informed decisions around retail savings and investment product choices.

Management + admin + other = 1.5% + 0.5% + 3% = 5%

Distribution waterfall = (20% - 6%) * 20% = 2.8%

Advisor = $((1 + 1.5\%) * (1 + 5\%)^9)^{(1/10)} - 1 = 0.6\%$

Thus EAC = 5% + 2.8% + 0.6% = 8.4%

[3]

iv)

a)

- Economic Survey 2021-22 estimates that the Indian economy (GDP) may grow by 9.2 per cent in real terms in 2021-22 (as per first advanced estimates) subsequent to a contraction of 7.3 per cent in 2020-21.
- GDP is projected to grow by 8- 8.5 per cent in real terms in 2022-23.
- The year ahead is poised for a pickup in private sector investment with the financial system in a good position to provide support for the economy's revival.
- A combination of high foreign exchange reserves, sustained foreign direct investment, and rising export earnings will provide an adequate buffer against possible global liquidity tapering in 2022-23.
- The economic impact of the "second wave" was much smaller than that during the full lockdown phase in 2020-21, though the health impact was more severe.
- The economic impact of "third wave" was minimal given the level of preparedness at all levels, vaccination drive and milder variant.
- Government of India's unique response comprised of safety-nets to cushion the impact on vulnerable sections of society and the business sector, significant increase in capital expenditure to spur growth and supply-side reforms for a sustained long-term expansion.
- The ongoing war between the Russia and Ukraine has added another layer of uncertainty in the market. Its direct impact is visible in the fast-spiraling crude oil prices which has already risen by ~39% since start of this year and currently trading above USD 110 per barrel.

- Russia plays an outsized role in the global oil market. It is the largest exporter of oil in the world accounting for almost 8% of total world oil supply. Thus, a prolonged war or stricter economic sanctions on Russian energy and financial sectors could put significant upward pressure on crude oil prices
- The government's flexible and multi-layered response is partly based on an "Agile" framework that uses feedback-loops, and the use of eighty High-Frequency Indicators (HFIs) in an environment of extreme uncertainty.
- Gross Tax Revenue registers a growth of over 50 per cent from April to November 2021 in YoY terms. This performance is strong compared to pre-pandemic levels of 2019-2020 also.
- India's merchandise exports and imports rebounded strongly and surpassed pre-COVID levels during the current financial year. There was a significant pickup in net services with both receipts and payments crossing the pre-pandemic levels, despite weak tourism revenues.
- As of end-November 2021, India was the fourth-largest forex reserves holder in the world after China, Japan and Switzerland.
- Economic Survey 2021-22 notes that the liquidity in the system remained in surplus.
 - Repo rate was maintained at 4 per cent in 2021-22.
 - RBI undertook various measures such as G-Sec Acquisition Programme and Special Long-Term Repo Operations to provide further liquidity.
- The average headline CPI-Combined inflation moderated to 5.2 per cent in 2021-22 (April-December) from 6.6 per cent in the corresponding period of 2020-21.
- The decline in retail inflation was led by the easing of food inflation.
- Food inflation averaged at a low of 2.9 per cent in 2021-22 (April to December) as against 9.1 per cent in the corresponding period last year.
 - Effective supply-side management kept prices of most essential commodities under control during the year.
 - Proactive measures were taken to contain the price rise in pulses and edible oils.
 - Reduction in central excise and subsequent cuts in Value Added Tax by most States helped ease petrol and diesel prices.
- Rupee has fallen in value significantly against the US dollar
- Large Government GSEC issuance as part of policy to stimulate the economy ; Increasing issuance of government debt may crowd out private investment debt

Key concerns:

- The Russia/Ukraine conflict continues to escalate.
- Inflation rises in a disorderly way, negatively affecting asset values.
- Policymakers remove accommodation too rapidly, undermining the global economic expansion.
- COVID-19 cases increase, including any new variants like omicron+delta.

[5]

b)

US –

Equity Market

- S&P 500 performance for February (-3.14%) followed January (-5.67%) on the downside bringing its YTD return to -8.23%.
- While the market continued to adjust (reallocate) itself for an expected slower (and more expensive) economy.
- Ukraine situation dominated the moral headlines and wreaked havoc on the market, with a knee-jerk reaction to an expected event, it was the economy that ruled the market.
- And for February, the economy was defined by increasing inflation stats (7.5% CPI, 9.7% PPI, and 6.1% PCE) and oil's return to USD 100 (last seen in July 2014).
- Latest loss in S&P was blamed mostly on inflation, as it overtook COVID-19 on the concern list.

- For most of this month, global conflict replaced inflation (through the correction entry), but inflation (costs of materials, labor, pass-throughs, etc.) reappears as a main driving force, slightly beating out second place global conflict, with COVID-19 a distant third.

[Max 1]

Bond Market

- The 10-year U.S. Treasury yield spiked 81 basis points over the first quarter to end the period at 1.74%.
- However, as the economic recovery became bumpier and inflation continued to rise, the yield curve partly retraced the first quarter's steepening over the remainder of the year.
- Short-term yields modestly rose, while the 10-year yield declined to 1.52% by the end of 2021.
 - Fed Tightens Policy Amid Rising Inflation
 - Credit Investors Hunt for Yield
 - Strong Dollar Weighs Down Global Returns
 - Demand for Municipal bond continues
- Performance across fixed-income Morningstar Categories was mixed as more-credit-sensitive strategies outpaced most of the pack, while interest-rate-sensitive and non-U.S. dollar-denominated bond funds posted losses.
- High-yield municipal funds led the way, with an average return of 5.7%, while emerging-markets local-currency bond funds plunged 7.3% on average.
- The Morningstar U.S. Core Bond Index, a proxy for typical U.S. bond exposure, fell 1.6% for the year, posting its worst calendar-year return since the taper tantrum roiled fixed-income markets in 2013.

[Max 1.5]

India

Equity:

- New strains of Covid-19, vaccination coverage, the U.S. Federal Reserve's timeline for tapering monetary stimulus, inflation, work-from-home, and retail investor frenzy shaped the equity markets in 2021.
- India's stock benchmarks were the top performers among global peers as they scaled new records.
- That rubbed off on the primary market with Indian companies raising a record amount via initial public offerings.
- The markets, however, ended the year volatile, having retreated from their October peak.
- All the nine sectoral indices on the NSE rose in 2021. The NSE Nifty Metal Index rose the most followed by Nifty IT Index and Nifty Realty Index. Bank, FMCG and pharma indices gained the least, but those gains were also at least 10%, as in the case of Nifty FMCG.
- While foreign inflows drove the markets higher till February, domestic institutional investors came to the rescue after the second wave of Covid-19.
- With concerns over faster withdrawal of monetary stimulus and the Omicron strain, the DIIs kept the markets afloat, undeterred by the FII outflows.
- India's stock benchmarks were among the top performers among the global markets in 2021.

[Max 1]

Bonds:

- Economic Survey 2021-22 notes that the liquidity in the system remained in surplus.
- Repo rate was maintained at 4 per cent in 2021-22.
- RBI undertook various measures such as G-Sec Acquisition Programme and Special Long-Term Repo Operations to provide further liquidity.
- The economic shock of the pandemic has been weathered well by the commercial banking system:

- YoY Bank credit growth accelerated gradually in 2021-22; from 5.3 per cent in April 2021 to 9.2 per cent as of 31st December 2021.
- Bond market has been quite volatile in 2022 so far. The 10-year government bond yield which was hovering around 6.35% during December 2021, spiked to 6.95% (on February 4, 2022) in the aftermath of Union Budget; and is currently trading around 6.80%.
- The volatility in the bond market can be attributed to -
 - hawkish US FED policy projecting series of rate hikes and liquidity reduction
 - rising crude oil and other commodity prices
 - higher than expected fiscal deficit and market borrowings by the Central Government
- Interest rates/bond yields moved up. While the RBI held the policy rates unchanged and delivered an ultra-dovish policy statement indicating continuation of the current easy monetary policy regime for a longer period.
- Government borrowing has more than doubled in the last three years. The central government has pegged its gross market borrowing in FY 2022-23 at Rs. 14.95 trillion vs 10.46 trillion in FY22. To recall, the government borrowing was around Rs. 7.1 trillion in FY20 (before the pandemic) which jumped to Rs. 12.6 trillion in FY21. It is already much higher than natural market appetite.
- Notwithstanding this fact, the bond market was well behaved in the last two years due to –
 - (1) easy monetary policy and increasing liquidity surplus,
 - (2) large bond purchases by the RBI and
 - (3) increased demand from banks due to HTM relaxation

[Max 1.5]

[Max 5]

v)

- Performance fees are normally used to reduce conflict of interest between fund manager and client and align both parties' interests to the outcome of the fund.
- However, it may lead to inappropriate behaviour by investment managers if performance fee is structured incorrectly or ambiguously.
- There is no incentive for the manager to manage downside risk and the manager will seek to maximize the fund returns with little consideration for the potential downside risk. There is no downside participation by the manager if the fund underperforms; passing on the entire risk to the investors.
- Although the fund does not have an benchmark portfolio to assess relative performance, there is still misalignment of nature of return on underlying (USD) investment to that of flooring of WPI + 5%.
- The flooring benchmark should ideally have the same nature as that of the underlying investment, i.e. USD denominated and having linking to US inflation.
 - a US denominated index based on debt instruments may be used for benchmark. Some adjustment might be required (either adding or subtracting a certain %) to adjust for the risk of underlying debt instruments compared to the benchmark.
 - Whichever benchmark is chosen, it should be transparent and readily available; which in this case is fine as WPI index is regularly published by a government agency.
- There is no cap on sharing the returns if the fund outperforms.
 - We may introduce a cap on the sharing of outperformance; this will have some barrier towards risk seeking behaviour of manager
 - Alternatively, a rebate could be introduced as compensation if certain hurdles have not been achieved
- Introduce a high water mark in order to avoid the manager achieving fees twice e.g. if the fund grows from a certain year the manager will get the performance fee, if the fund falls to original level or below and again grows to a higher level, manager could potentially get similar performance fee for no significant benefit to investors.
- A period longer than one year for rolling return might be better (barring 1st and 2nd year). The shorter the period, the more performance could be attributed to luck and high chances to fund

- getting fee twice for same return if underlying investment show cyclical returns over rolling period.
- The one year measurement period is also short, considering the fact that underlying investments are expected to be of much longer term.
 - If WPI is 5%, this fee will be $2\% + \max((20\% - 10\%), 0) * 20\% = 4\%$. It is slightly higher than the 4% expected from management fees above. If WPI is 15%, this fee will be $2\% + \max((20\% - 15\%), 0) * 20\% = 3\%$. If WPI is above 20%, then fee is floored to 2%. Unless the return from underlying investment increase with benchmark, there is risk to the fund manager that the inflation may increase drastically leaving lower fee – another reason not to use the WPI benchmark.
 - Having alternative fee structures available to investors will be suit different types of investors and could be used as a marketing tool
 - Whether this fee is appropriate could be determined by comparing it to other similar funds in the market; if the fee is significantly higher than similar funds, than it will be difficult to sell the product and find enough subscribers.
 - However, 20% participation should not be seen as excessive as it is in line with market norms, particularly in comparison with hedge funds and fund of funds.
 - Compared to flat fees, performance fees are very complicated. Portfolio managers can have different interpretation of wording and can have legal implications.
 - Performance fees can be difficult to explain to stakeholders including Trustees and members. Flat fees are more transparent although significant.
 - Administration of performance fees is complicated and mistakes are easily made. Checks are required
 - Performance fees are unpredictable compared to flat fees
 - The charging structures does not include an initial fee, for a new start-up company it might be appropriate to have an initial fee in order to cover set-up costs.
 - The base fee will might encourage the manager to try and gather as much assets as possible without consideration for the fund capacity.
 - The base fee could be justified by the fact that managing private debt fund is lot more labour intensive and more due diligence is required
 - Flat fees are more in line with treating customer fairly (TCF), although TCF is an evolving concept and have not fully rooted in fund management industry in India.
 - Returns might be compromised if the fund grows too big, however since it is based in the US, capacity should not be an issue

[5+5=10]

[45 Marks]**Solution 2:**

- i) Professional and Amateur investors analyse investments differently. Their analysis and behavioural aspects may be different with regard to the following (1mark for each valid point with explanation)
1. Amateur investors have relatively short term investment horizon while professional investors will have long-term investment time frame.
 2. Amateur generally invests for earning quick money on listing, Professional finds best investment opportunities to compound wealth consistently and patiently.
 3. Amateur investors get panicky when stock market is volatile while Professional investors show patience and consistency.
 4. Amateur listen to random advice and takes their investment decision, Professionals have his full-time trustable advisor on whom he can rely and discuss to make the decision for his investments.
 5. Amateur tries to time the market based on sentiments, emotions, etc. Professionals stick to their strategy no matter what are the sentiments or their emotions.

6. Amateur read news, noise & information. Professionals read books, experiences, case studies, behaviours, and reports.
7. Amateur deviates from their initial strategy if someone suggests something new, Professionals know that they have started their strategy based on its fundamentals and have faith in it. So, they won't change it until they actually achieve their goals.
8. Amateur investors may be influenced by the discount while professional investors will consider discount vis-à-vis the value of the share.
9. Amateur might have a confirmation bias based on their own thought narrative. They will try to find out information about the company that they want believe.
10. Amateur investors may might think of this investment in a large government owned life insurance companies as secure without realising the management capability and long term prospects of the firm due to high competition in the market.

[10]

ii) 1 mark for each valid point with explanation

- Overconfidence: Overconfidence Bias results in false sense of understanding due to overconfidence. The researcher may have overconfidence regarding his research abilities and he may think that his research is perfect and his investments will not have any downside risks. His select past experience may have made added to his confidence.
- Regret Aversion: The avoidance of future possible regrets may lead to bias. In this case, the bias may be that is we do not invest in this company, this will be a case of lost opportunity. The bias may also come in due to fear of being wrong.
- Oversimplification - Investing in the company based on the fact that it is Government owned and that is the largest life insurance company is a sure shot success strategy without diving much deep into the financials.
- Status Quo - May prefer to keep the investment as is even if the investment is not appropriate.
- Confirmation Bias: Confirmation Bias results in interpreting any information obtained as confirming existing analysis.
- Hindsight Bias: Hindsight bias results in interpreting events that occurred in the past being more likely than they actually are.
- Conservatism Bias: Conservatism Bias results in lower expectation due to misplaced fear of over estimation.
- Anchoring and Adjustment bias: The initial analysis may be based on detailed review, however any subsequent advice based on new information without a comprehensive review results in anchoring and adjustment bias.
- Availability Bias: This results while assuming that the available information is they only available information.

[7]

iii) Risk Appetite depends on a number of factors. The investor in this context is an amateur high net worth Investor aged 65 years.

It can be assumed that this investor would have a diversified portfolio and must be taking investment advisory from a professional investment advisor.

I think he would have reasonable level of risk appetite and risk capability but investment strategy may depend on other conditions such as his assets, liability, financial goals.

The HNI investor might not want volatility in investment returns. In this context, a life insurance fund would be a better option as it offers diversification through investment in other private life insurance along with investment in the company he wants to invest.

Directly investing in the company may be little cost effective and may give him better control on the management of investments. However, investment through funds would be more desirable as the fund would monitor investments actively and reallocate investments with changing scenario.

There may be some exit loads in case of funds.

Equity returns and mutual fund return might be subject to different taxation rules.

[5]

- iv) Lifecycle investments are investment strategy which are aligned to the risk appetite and risk capability of an individual. Generally, investors change their investment allocation from high risk investment to low risk investment as they grow old.

It is based on the concept that investors may have aggressive investment approach initially with high risk high return paradigm and can move to less risky assets with conservative approach to preserve their capital.

My advice to the client would be to invest in equity fund of an Asset management Company who has investments in other insurance companies as well. This advice would also depend on the client's liability, lifestyle, financial goals etc.

[3]

- v) (max 3 mark for each category of performance measure with explanation)

Absolute Performance Measurement. – This can be done for a fund as a whole or/and for individual asset categories or individual investments. The measurement tool could be Internal rate of return or any other absolute measure.

Relative Performance measurement - Relative to –

- An investment index
- A notional benchmark fund and
- A specified universe of other funds

The performance measurement process begins with the selection of an appropriate benchmark that will be subsequently used to assess the performance of a portfolio

Excess return is the difference between a portfolio's return and its benchmark's return. Excess return can be calculated arithmetically or geometrically:

Performing a Risk Analysis

Risk analysis is important for those who are responsible for both managing and controlling the portfolio's risk. Risk managers view risk positively. Basic risk measures can be divided into three categories:

- Absolute risk measures, such as standard deviation, the Sharpe ratio, and M2.

- Relative risk measures, such as tracking error and the information ratio.
- Regression, which measures the alpha, beta, and standard error of the portfolio's return.

Performance Attribution Analysis

Performance attribution quantifies the relationship between a portfolio's excess returns and the active decisions of the portfolio manager. In other words, it relates the excess returns of the portfolio (both positive and negative) to the active investment decisions of its manager.

There are three main types of attribution:

- Returns-based attribution, which uses factor analysis.
- Holdings-based attribution, which is calculated on a periodic basis and uses holdings data.
- Transactions-based attribution, which is calculated from holdings and transactions data

Against Liability:

- Change in assets with respect to liabilities.
- Is there any change in liability nature, term, currency and certainty from initial allocation.

[10]

[35 Marks]

Solution 3:

i) (1 mark for each valid point with explanation)

- CBDC is the legal tender issued by a central bank in a digital form.
- CBDC is the same as currency issued by a central bank but takes a different form than paper.
- It is sovereign currency in an electronic form and it would appear as liability (currency in circulation) on a central bank's balance sheet
- CBDCs is exchangeable at par with cash.
- CBDC is a digital or virtual currency but it is not comparable to the private virtual currencies that have mushroomed over the last decade
- It is the same as a fiat currency and is exchangeable one-to-one with the fiat currency. Only its form is different.
- Transactions are processed and recorded on a blockchain, which is a public, distributed ledger.

[5]

ii) The rationale includes the following (1mark for each valid point with explanation)

- We may seek to popularize a more acceptable electronic form of currency to improve economic and financial inclusion
- To make issuance more efficient particularly in the Jurisdictions with significant physical cash usage
- To meet the public's increasing need for digital currencies
- The increasing demand is manifested in the increasing use of private virtual currencies. This will avoid the potentially more damaging consequences of such private currencies.
- To reduce settlement risk in the financial system
- This will potentially enable a more real-time globalization of payment systems
- This will cost-effective payment systems

[5]

iii) Introduction of CBDC has the potential to provide significant benefits, such as

- reduced dependency on cash,
- higher seigniorage due to lower transaction costs
- reduced settlement risk.
- more robust, efficient, trusted, regulated and legal tender-based payments option.

There are associated risks, no doubt, but they need to be carefully evaluated against the potential benefits.

- India's high currency to GDP ratio holds out a great benefit of CBDCs. The cash usage can be replaced by CBDCs, the cost of printing, transporting, storing and distributing currency can be reduced.
- CBDCs can cause a reduction in the transaction demand for bank deposits. Since transactions in CBDCs reduce settlement risk as well, they reduce the liquidity needs for settlement of transactions (such as intra-day liquidity).
- At the same time reduced disintermediation of banks carries its own risks. If banks begin to lose deposits over time, their ability for credit creation gets constrained.
- Since central banks cannot provide credit to the private sector, the impact on the role of bank credit needs to be well understood
- Availability of CBDC makes it easy for depositors to withdraw balances if there is stress on any bank. Flight of deposits can be much faster compared to cash withdrawal.
- CBDC ecosystems may be at similar risk for cyber-attacks as the current payment systems are exposed to.
- Further, in countries with lower financial literacy levels, the increase in digital payment related frauds may also spread to CBDCs.
- Ensuring high standards of cybersecurity and parallel efforts on financial literacy is therefore essential for any country dealing with CBDC.
- Absorption of CBDCs in the economy is also subject to technology preparedness.
- The creation of population scale digital currency system is contingent upon evolution of high speed internet and telecommunication networks and ensuring the wider reach of appropriate technology to the general public for storing and transacting in CBDCs.

[10]

[20 Marks]
